

Series Sheath Capacitance of a Plasma-Immersed Dipole in Three-Dimensional Fluid FDTD: Volumetric Jacket Versus Analytic Coaxial Loading

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Key points

- A prescribed vacuum jacket around a dipole in 3D fluid FDTD acts as series sheath capacitance, not a shunt feed capacitor.
- Low-frequency CW shows $\text{Im}\{Z\}$ zero-crossings falling with sheath width, opposite to a dielectric-unloading upward-shift scoreboard.
- Band-averaged $C_{\text{eff}}(S_d)$ tracks coaxial analytic $C_{\text{sh}}(S_d)$ in trend with a nearly constant scale factor near 0.3 for $S_d = 2$ and 10.

Abstract

We quantify how a prescribed volumetric vacuum sheath of width S_d cells loads the feed impedance of a thin-wire dipole in a three-dimensional plasma-fluid finite-difference time-domain (PF-FDTD) solver. After attaching the sheath to the conductor (PEC-seeded density depletion), continuous-wave (CW) phasor measurements at plasma frequency $f_p = 2$ MHz show strong S_d dependence of $Z(f)$. Using the first $\text{Im}\{Z\}$ positive-to-negative crossing as a series-resonance marker yields $f_{\text{res}}(0) \approx 1.85$ MHz, $f_{\text{res}}(2) \approx 0.66$ MHz, and $f_{\text{res}}(10) \leq 0.50$ MHz: the crossing moves downward with sheath thickness. We interpret this as series sheath capacitance $1/(j\omega C_{\text{sh}})$ dominating dielectric unloading. Defining an effective capacitance $C_{\text{eff}} = -1/(\omega \text{Im}\{Z\})$ on capacitive branches and averaging over 0.8–1.2 MHz gives $C_{\text{eff}}(2) \approx 7.4$ pF and $C_{\text{eff}}(10) \approx 4.3$ pF. These values decrease with S_d in the same sense as a coaxial analytic sheath $C_{\text{sh}} = 2\pi\epsilon_0 L / \ln(1 + S_d \Delta x / r_{\text{eff}})$ with Holland-like $r_{\text{eff}} = 0.23\Delta x$, with $C_{\text{eff}}/C_{\text{sh}} \approx 0.30$ at both thicknesses. The work provides a quantitative bridge between volumetric FDTD sheaths and Balmain/Liu-type series gap models used for plasma impedance probes, and clarifies why an upward $f_{\text{res}}(S_d)$ target inspired by transmitting-antenna sheath literature is the wrong scoreboard for a static prescribed jacket.

Plain Language Summary

Radio antennas in space plasmas sit inside a thin empty “sheath” next to the metal. We simulated that empty layer as a jacket of controlled thickness around a dipole in a 3-D plasma wave code. Making the jacket thicker made the antenna look more capacitive and moved its resonance to lower frequency, which matches treating the jacket as a series capacitor. The effective capacitance we extracted from the simulations decreases with jacket thickness in the same way as a simple coaxial capacitor formula, up to a nearly constant scale factor. This helps connect detailed wave simulations to the circuit models used to interpret plasma impedance probes.

1 Introduction

A conducting antenna immersed in plasma is surrounded by a depleted sheath that can dominate low-frequency impedance. Classical short-dipole theory in a cold magnetoplasma (Balmain(1964), almain(1964); Balmain(1969), almain(1969)) provides the homogeneous-medium baseline; later Balmain-era and contemporary treatments add ion sheaths, in-

insulating jackets, and sheath-wave physics as series gap loading between metal and exterior plasma. Modern plasma impedance probe (PIP) retrievals often write an explicit coaxial sheath capacitor in series with a Balmain-like plasma term (Liu et al.(), iu et al.()), so that the phase-zero frequency lies below bulk plasma features.

High-voltage transmitting antennas raise a related but distinct problem: Song and coworkers modeled an oscillating ion-sheath radius about a bare dipole (Song et al.(2007), ong et al.(2007)), and Tu et al. used one-dimensional particle-in-cell (PIC) simulations to refine sheath structure and reactance (Tu et al.(2008), u et al.(2008)). Those studies motivate the statement that “sheath changes antenna impedance,” but they do not prescribe a fixed vacuum jacket in three-dimensional fluid FDTD, nor do they require that an $\text{Im}\{Z\} = 0$ marker climb with sheath thickness.

This paper addresses a narrower, quantitative question suited to the PF-FDTD solver used here: does a *prescribed* volumetric vacuum jacket of width $S_d\Delta x$ behave as a *series* sheath capacitor whose effective $C_{\text{eff}}(S_d)$ tracks coaxial analytic expectations? Campaign diagnostics that established conductor-attached sheath coupling and the downward $f_{\text{res}}(S_d)$ trend are summarized only as needed; the methodological path (including a pre-fix seeding bug) is documented elsewhere (Paper0 campaign(), aper0 campaign()). Success is defined by C_{eff} versus coax and multi-marker reporting of $Z(f)$, not by an upward Im-zero shift with S_d .

2 Methods

2.1 PF-FDTD solver and antenna geometry

PF-FDTD advances Maxwell’s equations on a staggered Yee grid coupled to multi-species fluid continuity and momentum equations. Plasma current enters Ampere’s law; feed observables are voltage $V(t)$ and current $I(t)$ at a hard E_z source, with

$$Z(f) = \frac{V(f)}{I(f)} \quad (1)$$

from steady-state CW phasors on the last 50% of each `.vc` record.

All runs use a $70 \times 70 \times 65$ grid with $\Delta x = 0.04$ m and a z -directed thin-wire dipole from grid indices (35, 35, 20) to (35, 35, 45) (length $L = 1.00$ m) with the feed at (35, 35, 33). Unless noted, $f_p = 2$ MHz, collisions are weak, and the background magnetic field is zero.

2.2 Prescribed volumetric sheath

Ambient density is initialized uniformly, then depleted in a shell of width S_d cells about PEC cells ($\text{ER}_x\text{ER}_y\text{ER}_z = 0$):

$$0 < d \leq S_d \quad \Rightarrow \quad N_0 \rightarrow N_0 \cdot 10^{-6}, \quad (2)$$

so plasma current is negligible and the jacket behaves as vacuum. Sheath seeding from the plasma-on mask (rather than PEC) previously placed depletion at the domain boundary and made Z independent of S_d ; all results here use the PEC-seeded implementation (Paper0 campaign(), aper0 campaign()).

2.3 CW campaigns

Two CW grids share the same geometry and f_p :

Dense band $S_d \in \{0, 2, 4, 6, 8, 10\}$, $f = 1.50\text{--}2.30$ MHz (100 kHz and selected half steps; 66 cases).

Low-frequency extension $S_d \in \{0, 2, 10\}$, $f = 0.50\text{--}1.50$ MHz (100 kHz steps), to locate crossings for $S_d \geq 2$.

Series resonance f_{res} is the first $\text{Im}\{Z\}$ positive-to-negative zero crossing by linear interpolation. We also retain $|Z|$ structure as a second marker and do not equate phase-zero with magnitude extrema.

2.4 Effective capacitance and coaxial reference

Where $\text{Im}\{Z\} < 0$, define

$$C_{\text{eff}}(f) = -\frac{1}{\omega \text{Im}\{Z(f)\}}, \quad \omega = 2\pi f. \quad (3)$$

This C_{eff} is an *effective* feed capacitance of the full antenna–sheath–plasma system, not a pure vacuum gap to an ideal plasma electrode. Band averages use 0.8–1.2 MHz (capacitive points only), away from the narrow inductive island of $S_d = 2$ near 0.60 MHz.

The coaxial analytic sheath for a cylinder of length L and effective radius r_{eff} is

$$C_{\text{sh}}^{(\text{coax})} = \frac{2\pi\epsilon_0 L}{\ln(1 + S_d \Delta x / r_{\text{eff}})}. \quad (4)$$

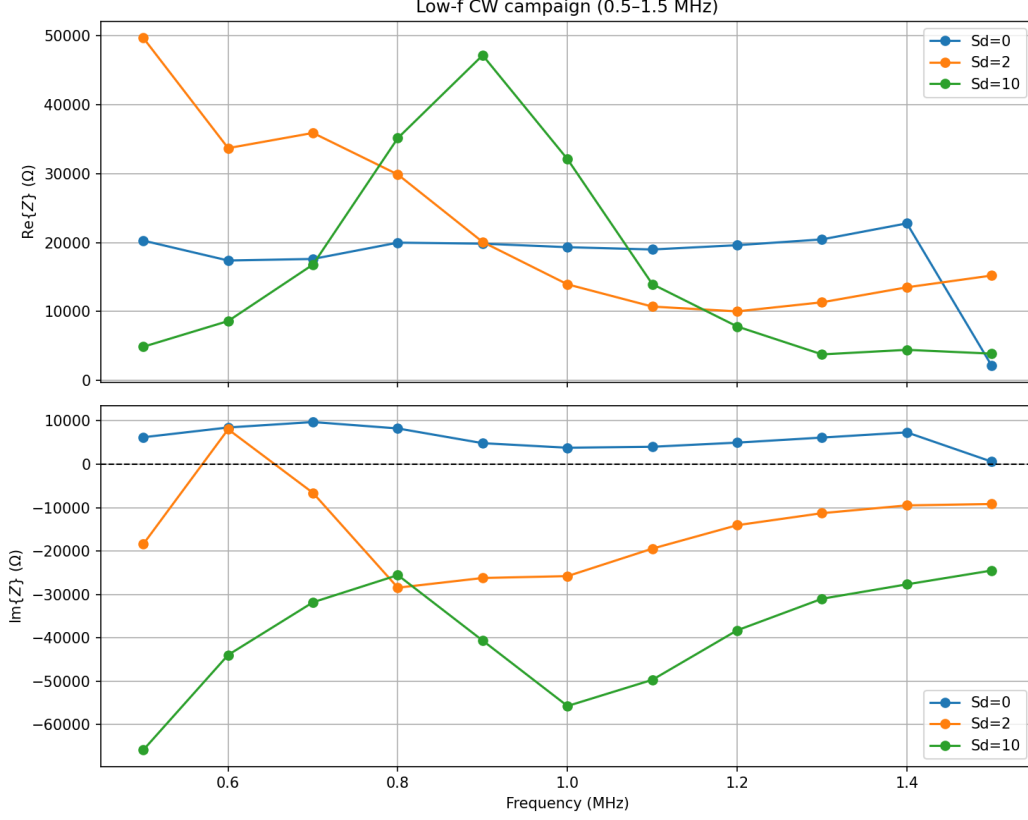


Figure 1. Low-frequency CW phasor impedance for $S_d = 0, 2, 10$ ($f_p = 2$ MHz). Top: $\Re\{Z\}$; bottom: $\Im\{Z\}$. The outlier at $S_d = 0$, 1.50 MHz is an incomplete restart file and is discarded for physics.

We take $r_{\text{eff}} = 0.23\Delta x$ as a Holland-type thin-wire estimate on the Yee lattice (Holland and Simpson(1981), olland and Simpson(1981)); sensitivity to r_{eff} is discussed in Section 4.

3 Results

3.1 Impedance spectra and sheath coupling

Figure 1 shows low-frequency $\Re\{Z\}$ and $\Im\{Z\}$ for $S_d = 0, 2, 10$. Curves remain strongly separated ($|\Delta Z|/|Z_0|$ typically 80–350%), confirming that the volumetric jacket enters the feed. $S_d = 0$ stays inductive through 0.50–1.50 MHz; $S_d = 10$ is capacitive at every low-frequency point; $S_d = 2$ is capacitive except for a brief inductive island at 0.60 MHz.

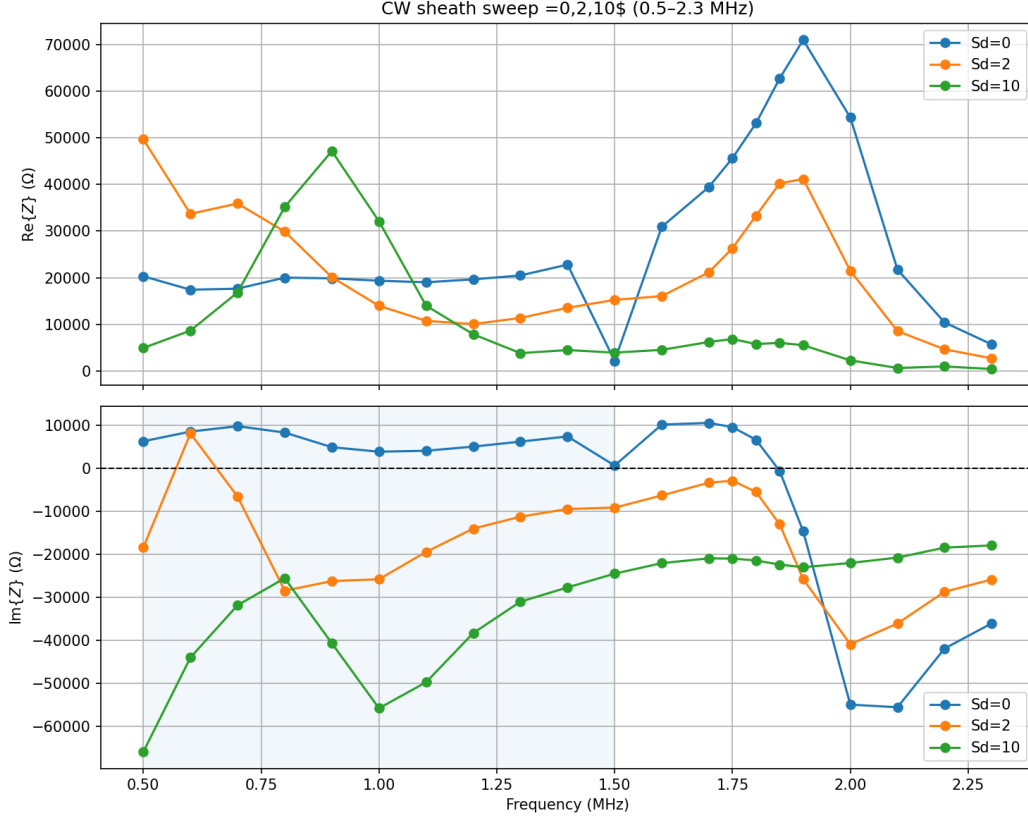


Figure 2. Combined CW impedance for $S_d = 0, 2, 10$ over 0.50–2.30 MHz. The shaded region marks the low-frequency extension.

Figure 2 combines the low-frequency extension with the dense high band. For $S_d \geq 2$, $\text{Im}\{Z\}$ remains negative through 1.50–2.30 MHz; only $S_d = 0$ shows a high-band inductive-to-capacitive transition near 1.85 MHz.

3.2 Resonance markers versus sheath width

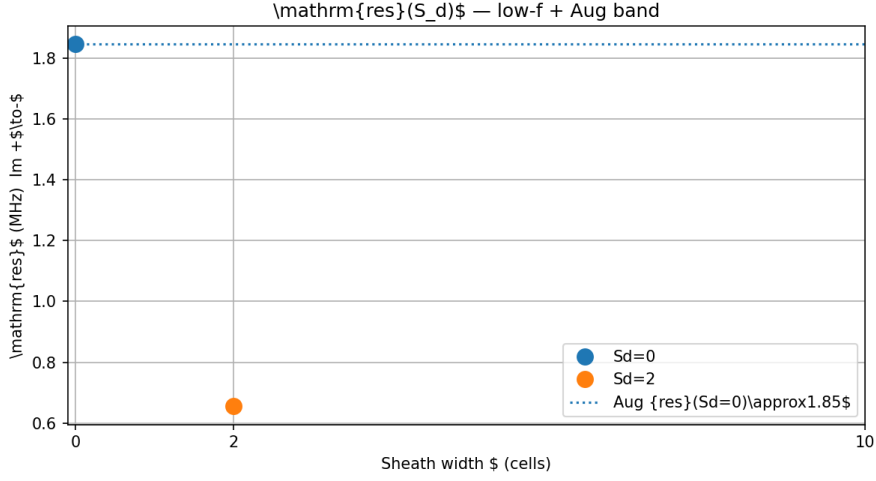
Table 1 and Figure 3 summarize $\text{Im}\{Z\}$ positive-to-negative crossings. Widening the jacket moves f_{res} *downward*, opposite to a dielectric-unloading upward-shift expectation for static S_d .

3.3 Effective capacitance versus coaxial sheath

Figure 4 shows $C_{\text{eff}}(f)$ from equation (3) on capacitive branches. Band means over 0.8–1.2 MHz are listed in Table 2 and compared with equation (4) in Figure 5.

Table 1. Series resonance from $\text{Im}\{Z\}$ + to – crossings

S_d	f_{res}	Note
0	1.846 MHz	August high band; low-frequency band all inductive
2	0.655 MHz	Also – to + at 0.570 MHz (inductive island)
10	≤ 0.50 MHz	All $\text{Im}\{Z\} < 0$ in 0.50–1.50 MHz

**Figure 3.** Resonance frequency $f_{\text{res}}(S_d)$ from $\text{Im}\{Z\}$ positive-to-negative crossings. $S_d = 10$ lies at or below the lower band edge.**Table 2.** Band-averaged C_{eff} (0.8–1.2 MHz) versus coaxial C_{sh}

S_d	t_{sh} (m)	C_{eff} (pF)	$C_{\text{sh}}^{(\text{coax})}$ (pF)	Ratio
2	0.08	7.35 ± 1.12	24.49	0.30
10	0.40	4.27 ± 1.83	14.66	0.29

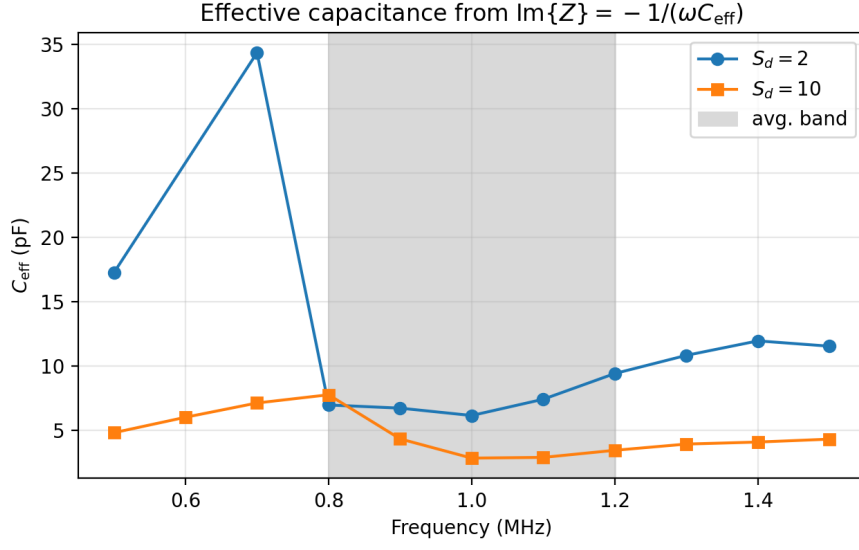


Figure 4. Effective capacitance $C_{\text{eff}}(f)$ for $S_d = 2$ and 10. Shaded band marks the averaging window used in Table 2.

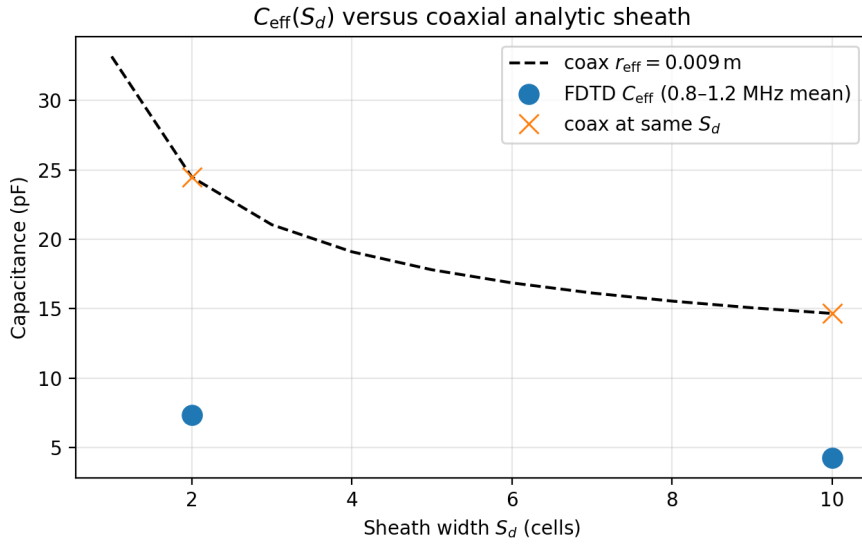


Figure 5. $C_{\text{eff}}(S_d)$ (symbols) compared with coaxial analytic $C_{\text{sh}}(S_d)$ for $r_{\text{eff}} = 0.23\Delta x$ (dashed). Crosses mark coax evaluated at the simulated S_d .

Both thicknesses show $C_{\text{eff}} < C_{\text{sh}}^{(\text{coax})}$ by a nearly common factor ≈ 0.30 . Absolute scale therefore depends on r_{eff} , staircasing, and residual plasma reactance inside C_{eff} , but the *trend* with S_d matches series coaxial loading: thicker jackets reduce capacitance and strengthen capacitive reactance at fixed frequency.

4 Discussion

4.1 Series versus parallel topology

The depleted shell sits between the conductor and bulk plasma. Feed current must cross the gap by displacement current, so sheath loading appears in series with the plasma-loaded antenna impedance,

$$Z_{\text{in}}(f) \approx Z_{\text{pl}}(f) + \frac{1}{j\omega C_{\text{sh}}(S_d)}, \quad (5)$$

not as a shunt capacitor across the terminals. Parallel-plate estimates $C \sim \varepsilon_0 A / (S_d \Delta x)$ are pedagogical magnitude guides for that *series* gap; they do not imply parallel circuit topology. Thin-wire geometry favors the logarithmic coax form (equation (4)).

4.2 Why f_{res} falls with S_d

Thickening the jacket simultaneously (i) removes plasma dielectric near the wire (favoring an upward shift toward free space) and (ii) inserts series C_{sh} (favoring a downward phase-zero shift, as in PIP $f_{\phi 0}$ arguments). Under the $\text{Im}\{Z\}$ crossing metric on this grid, series capacitance wins. On $\Delta x = 0.04 \text{ m}$, $S_d = 2$ is already an 8 cm vacuum jacket—a strong capacitive load relative to a physical Debye sheath—which amplifies series- C domination.

4.3 Relation to Tu et al. and Song et al.

Tu (Tu et al.(2008), u et al.(2008)) and Song (Song et al.(2007), ong et al.(2007)) address high-voltage transmitting dipoles with self-consistent, time-varying sheaths. The present runs prescribe a static step jacket in fluid FDTD. We therefore do not claim Tu dynamic fidelity. The appropriate static bridge is $C_{\text{sh}}(r_s)$ (and later, in follow-on work, prescribed $\dot{r}_s$), not an upward Im -zero scoreboard.

4.4 Limitations

1. Free-space f_{res} on the same dipole remains an open upper anchor for the unloading narrative.
2. C_{eff} mixes sheath and plasma reactance; a pure gap extraction would require additional modeling (e.g., vacuum reference or circuit de-embedding).
3. Absolute coax comparison depends on r_{eff} ; the reported ratio ≈ 0.3 is for $r_{\text{eff}} = 0.23\Delta x$. Larger r_{eff} raises C_{sh} further; smaller r_{eff} improves absolute agreement but does not change the shared S_d trend.
4. Campaigns here are effectively unmagnetized; PIP upper-hybrid markers require $B \neq 0$.
5. Only $S_d = 2$ and 10 enter the low-frequency C_{eff} table; intermediate widths were capacitive in the dense band but were not re-run below 1.5 MHz.

5 Conclusions

A prescribed volumetric vacuum sheath in PF-FDTD loads a plasma-immersed dipole as a **series** capacitor. CW measurements show $\text{Im}\{Z\}$ zero-crossings decreasing with S_d , and band-averaged $C_{\text{eff}}(S_d)$ decreasing in parallel with coaxial analytic $C_{\text{sh}}(S_d)$ at nearly constant scale factor ≈ 0.3 for $S_d = 2$ and 10. These results validate series-gap loading as the correct FDTD interpretation of a static jacket and provide the quantitative baseline needed before prescribing time-varying sheath boundaries or self-consistent charging.

Acknowledgments

The authors thank collaborators on the PF-FDTD sheath campaign for discussions of coupling diagnostics and impedance markers.

Open Research

Simulation outputs supporting the figures are archived under the campaign results tree (`results/sheath_cw_tu/`) and derived tables in `paper1_static_sheath_capacitance/analysis/data/`. Analysis scripts include `make_ceff_products.py`. Solver source is available in the colab PF-FDTD repository (branch with PEC-seeded `ApplySheath`). Update this section with a persistent DOI/repository link before submission.

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